

Round 3: Re-solving the Equilibrium

What is changing, and what is not

The theory has been sharpened since Round 2, and one consequence is that we now have exact formulas for everything the equilibrium solver computes. That is good news and it makes this round short.

Nothing about the stock-return work changes. Parts 1, 2, 3, 5, 6, and 7 are final as delivered. The comovement results, the clustered standard errors, the 200-replication Monte Carlo, the power curve, and the event-day analysis all stand. Please do not re-run them. One point of reassurance: Round 1 drew each firm's usage δ_i from a distribution, and the theory at that time implied every firm should choose the same usage. That inconsistency has been resolved in the theory's favour—firms now differ in how expensive AI is to integrate, which makes usage differ across firms exactly as you simulated it. Your setup was right; the model has caught up.

The equilibrium work (Parts 4 and 8) needs re-solving. Not because the economics changed, but because the solution method was too coarse. You reallocated firms in 5% steps, so the market share could only land on multiples of 5%. The true equilibrium share in the frontier experiment is 78.4%, and the solver reported 80%. That rounding is responsible for about a third of the headline number we reported, so it has to go.

New terms

Quality ratio

Previously we described the leading vendor's advantage as a *gap* (e.g. 0.20 versus 0.18). We now use the *ratio* ($0.20/0.18 = 1.11$), because the theory says the market tips at a ratio and the gap version depends on units.

Crowd charge steepness k

The coefficient in the crowding charge, previously written as the number 0.10. Keep $k = 0.10$.

Cost curvature χ

The coefficient in the adoption cost $\frac{\chi}{2}\delta^2$. Keep $\chi = 1$.

Tipping threshold

The theory says both vendors survive if and only if the quality ratio is below $1 + k/\chi$. With our numbers that is exactly 1.10.

HHI (Herfindahl index)

A standard concentration measure. If vendor v has share s_v of total delegation, $\text{HHI} = \sum_v s_v^2$. It equals 0.5 when two vendors split the market evenly and 1 when one vendor has everything.

Ground rules

Same as before: seeded, one config file, one script per exhibit. New this round: **every equilibrium number you produce must be checked against the formula given below**, and any disagreement beyond the eighth decimal place is a bug to be found, not a result to be reported.

1 Part 9: Solve the equilibrium exactly (half a day)

Replace the 5%-reallocation loop with a direct solution. For quality levels Δ_a (leader) and Δ_b (rival), with $k = 0.10$ and $\chi = 1$:

1. Compute the common net margin $\bar{m} = (\Delta_a + \Delta_b)/(2 + k/\chi)$.
2. Compute vendor masses $D_a = (\Delta_a - \bar{m})/k$ and $D_b = (\Delta_b - \bar{m})/k$.
3. **If $D_b \leq 0$, the market has tipped.** Use instead $D_a = \Delta_a/(\chi + k)$ and $D_b = 0$.
4. Total delegation is $\delta^{\text{tot}} = D_a + D_b$; shares are $s_v = D_v/\delta^{\text{tot}}$; $\text{HHI} = s_a^2 + s_b^2$.
5. Systematic risk is $0.2 \times (\delta^{\text{tot}})^2 \times \text{HHI}$.

Keep the iterative solver as a check. Re-run your Round-1 best-response loop, but let the share move continuously (any step size that converges, or a bisection on the share) rather than in 5% jumps. It should reproduce the formula above to at least eight decimals.

Report: a short table comparing formula versus iterative solver at three quality ratios (one interior, one at the boundary, one tipped).

What you should see: exact agreement. If the iterative solver disagrees, the usual cause is applying the crowd charge with the previous iteration's crowd size instead of the current one.

2 Part 10: Redo the sweep and the frontier experiment (half a day)

1. **The sweep.** Hold $\Delta_b = 0.18$ and vary the quality ratio Δ_a/Δ_b from 1.00 to 1.28 in at least 100 steps. For each, record the leader's share, total delegation, HHI, and systematic risk.
2. **The frontier experiment.** Recompute the interior comparative static: Δ_a from 0.190 to 0.195, with $\Delta_b = 0.18$ throughout. Report the leader's share, total delegation, HHI, and systematic risk before and after, and—this is new—report the percentage change in each of the three components separately.

Report: Figure—leader's share against the quality ratio, with a vertical line at the tipping threshold 1.10; Figure—systematic risk against the quality ratio on the same horizontal axis; Table—the frontier experiment, before and after, with percentage changes.

What you should see. The share should rise smoothly from 50% at parity to about 95% just below the threshold, then jump to 100%—no staircase. Systematic risk should rise throughout the interior region, roughly doubling before any tipping occurs. The frontier experiment should give the leader's share moving from 78.4% to 92.0% and systematic risk rising about 32.5%, of which roughly 2.7 percentage points come from higher total delegation and 29 percentage points from higher concentration. These target values are given so you can confirm the code; if your numbers differ, the code is wrong, not the targets.

3 Part 11: Verify the risk decomposition (one hour)

The theory claims systematic risk factors exactly into an adoption piece and a concentration piece:

$$\text{systematic risk} = 0.2 \times \underbrace{(\delta^{\text{tot}})^2}_{\text{adoption}} \times \underbrace{\text{HHI}}_{\text{concentration}}.$$

For every point in the Part-10 sweep, compute the two pieces separately, multiply them, and compare to the systematic risk computed directly.

Report: one line stating the largest absolute discrepancy across the sweep.

What you should see: agreement to machine precision. This is an identity, not an approximation, so any discrepancy is arithmetic error.

4 Part 12: The survival region (half a day, new exhibit)

This replaces the old convergence-path figure, which showed only that the solver worked.

The theory says both vendors survive when the quality ratio is below $1 + k/\chi$, where $k = \gamma\theta\sigma_A^2$. Two ingredients therefore determine how much vendor diversity the market can sustain: how correlated AI errors are (θ) and how costly AI is to integrate (χ). Map that region.

1. Fix $\gamma = 0.5$ and $\sigma_A^2 = 1$, so $k = 0.5\theta$.
2. Over a grid of $\theta \in [0.01, 0.40]$ and $\chi \in [0.5, 4.0]$, compute the threshold $1 + k/\chi$.
3. Plot it as a contour or heat map, with θ on one axis and χ on the other, and contour lines at thresholds of 1.01, 1.02, 1.05, 1.10, 1.20, and 1.40.
4. Mark our calibration point ($\theta = 0.2$, $\chi = 1$, giving a threshold of 1.10).

Report: the contour figure, plus a small table of the threshold at $\theta \in \{0.05, 0.10, 0.20, 0.40\}$ crossed with $\chi \in \{0.5, 1, 2, 4\}$.

What you should see: thresholds ranging from about 1.006 (weakly correlated errors, expensive integration—almost no diversity survives) to 1.40 (strongly correlated errors, cheap integration—a leader can be 40% better and a rival still survives). The economic reading, which is the point of the figure: vendor diversity is most fragile exactly where AI errors are least correlated, and most robust where they are most correlated.

Deliverables summary

Part	Deliverable	Purpose
9	Formula vs. solver table	Certify the exact solution
10	Share-vs-ratio figure	Tipping at 1.10, no staircase
10	Risk-vs-ratio figure	Risk doubles before tipping
10	Frontier table with components	Corrected headline number
11	One-line identity check	Confirms the decomposition
12	Survival-region contour + table	Supports the new prediction

Suggested pace: two working days. Retired exhibits: the Round-1 convergence-path figure and the old gap-based sweep figure are superseded by Parts 10 and 12. Please also send the corrected equilibrium script separately, since the formulas in Part 9 are now the reference implementation. As before, close with a one-page memo in words describing what changed and by how much.